

CHEM 101 Topic 6: Names and Formulas — Problem Set

Instructions: Complete the following problems using the analysis method practiced in your guided assignment. Show all your work, including your charge balancing for ionic compounds, prefix identification for molecular compounds, and formula derivations for acids and hydrates.

Section 1: Naming Ionic Compounds (Fixed-Charge Metals)

Problem 1: Name the compound NaF.

Problem 2: Write the chemical formula for Calcium sulfide.

Problem 3: Write the chemical formula for Aluminum oxide.

Section 2: Variable-Charge Metals & Polyatomic Ions

Problem 4: Name the compound Na₂SO₄.

Problem 5: Name the compound FeCl₃.

Problem 6: Write the formula for Copper(II) nitrate.

Section 3: Binary Molecular Compounds (Nonmetals)

Problem 7: Name the compound SF₆.

Problem 8: Write the formula for Carbon tetrafluoride.

Problem 9: Name the compound Cl₂O₇.

Section 4: Acids, Salts, and Hydrates

Problem 10: Name the acid HBr(aq).

Problem 11: Name the acid H₂SO₃.

Problem 12: Name the compound BaCl₂ · 2H₂O.